End-to-End E-Commerce Analytics

A full-stack data solution — from 2 lakh+ raw data cleaning in Python, through SQL business queries, to a strategic Power BI dashboard — uncovering 8 high-impact insights that drive real business decisions.

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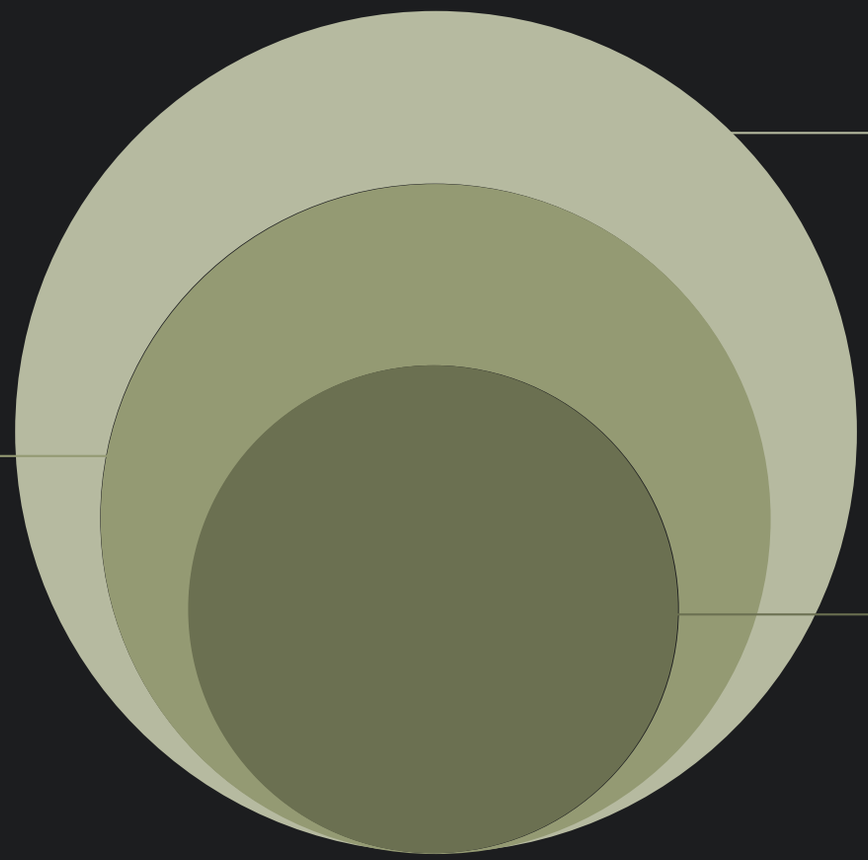
PROJECT ARCHITECTURE

Three-Layer Analytics Pipeline



Dashboard Layer

Power BI: KPIs, trends, decision support

SQL Layer

Business queries: revenue, segmentation, delivery

Data Layer

Python : cleaning, EDA, reliability

Each layer builds on the last: clean data enables accurate queries, and accurate queries power dashboards that drive decisions. This is the exact architecture used in production analytics teams.

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LAYER 1 — DATA FOUNDATION

Data Layer: Reliability Before Decisions

What Was Done

- Removed irrelevant columns to reduce noise
- Handled nulls & outliers to improve model accuracy
- Performed EDA — identified distributions, skewness, and patterns

Real-world meaning: Ensuring data reliability before decision-making is critical in production systems. Bad data leads to bad strategy.

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LAYER 2 & 3 — INTELLIGENCE

SQL Queries + Power BI Dashboard

41.16M

Total Revenue

Across all orders

99K

Customers

Active buyer base

99.4...

Orders

Processed successfully

3.99

Avg Rating

Customer satisfaction

SQL answered: revenue, geography, segmentation, delivery, and product quality — **Descriptive + Diagnostic Analytics**. Power BI transformed findings into **Business Intelligence** with time-series trends, state-level maps, and payment breakdowns.

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STRATEGIC INSIGHTS 1–4

Where the Business Is at Risk — and Why

🔥 Geographic Over-Concentration

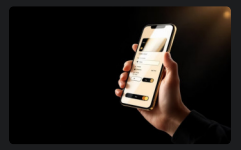
São Paulo = **80% of revenue**. Single-region dependency creates major risk. → **Expand into Tier-2/3 cities.**

⚡ Delivery Drives Ratings

Faster deliveries → ratings ~4+. Delivery is a **customer experience problem**, not just logistics. → **Prioritize same-day in metros.**

🏙️ High-Volume City Bottlenecks

Top-order cities face longer delivery times and delays. → **Open micro-fulfillment centers.**



💳 Payment Behavior Patterns

Credit cards = **~76%**. Customers prefer instant methods. → **Offer cashback, EMI, reduce friction.**

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STRATEGIC INSIGHTS 5–8

Hidden Revenue Opportunities

📦 Order Status Efficiency

97% delivered — strong ops, but even small cancellation rates at scale = big losses. Investigate root causes.

👑 Customer Segmentation (80/20 Rule)

Top customers drive disproportionate revenue. → **Build loyalty programs, personalized recommendations, subscriptions.**

📈 Seasonal Revenue Volatility

Growth from 2017→2018 with fluctuations. Demand is seasonal. → **Plan inventory pre-peak; discount in low months.**

🚨 Shipping Cost Leakage

High freight vs. product price kills conversion. → **Introduce free-shipping thresholds; optimize courier partnerships.**

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MY RECOMMENDED

The Business Story Is

> "I built an end-to-end e-commerce analytics solution — cleaning data in Python, extracting insights via SQL, and creating a Power BI dashboard with Dax formulas. I found revenue is concentrated in São Paulo, delivery time drives satisfaction, and shipping costs leak revenue. I recommended regional expansion, logistics optimization, and customer retention strategies."

🛠️ Tools

Python · SQL · Power BI

📊 Analytics Types

Descriptive · Diagnostic · BI

💡 Impact

8 strategic recommendations

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